

Chapter 8: Anonymization of datasets

8.1 Introduction:

The paper is **Majeed & Lee (2020)**, “**Anonymization Techniques for Privacy Preserving Data Publishing: A Comprehensive Survey.**” This is used as reference for this chapter preparation.

Health datasets contain sensitive information about individuals and may contain variables that can directly or indirectly identify a person. Therefore, before health data are shared with researchers, institutions, or other data users, privacy-preserving techniques may be applied to reduce the risk of identification and disclosure.

Anonymization can be understood as a transformation of an original dataset D into an anonymized dataset D' : $D \rightarrow D'$

The objective is to protect individual privacy while retaining sufficient information for legitimate analysis. This creates the traditional **privacy–utility trade-off**: stronger privacy protection may require greater modification of the original data and consequently reduce its analytical utility. The paper describes this problem as a central challenge in privacy-preserving data publishing (PPDP).

For this PhD, however, statistical or analytical utility alone is not sufficient. An anonymization procedure may preserve distributions, correlations, or predictive performance while altering causal relationships within the data. Therefore, an additional question is required:

Does anonymization preserve causal utility?

The central objective is therefore to extend the conventional privacy–utility framework towards:

Privacy ↔ Statistical Utility ↔ Causal Utility

8.2 Types of Variables in Health Data

Understanding the role of variables is an important first step in anonymization.

8.2.1 Direct Identifiers

Direct identifiers are variables that can uniquely identify an individual. Examples include:

- Name
- Patient identification number
- Social security number
- Email address
- Telephone number

These variables are generally removed before further anonymization. The paper describes the removal of directly identifiable information as the first stage of the anonymization process.

8.2.2 Quasi-identifiers

Quasi-identifiers (QIs) are variables that may not identify an individual on their own but may allow identification when combined with other variables or external information.

Examples include:

- Age
- Sex
- ZIP/postcode
- Occupation
- Date of admission

For example:

Age+Sex+ZIP+Occupation

may uniquely identify an individual when linked with external information.

The selection of QIs is therefore an important component of anonymization. The paper identifies QI selection as one of the major challenges in relational data anonymization because the privacy role of an attribute may depend on the context.

8.2.3 Sensitive Attributes

Sensitive attributes contain information that should not be disclosed about an individual.

Examples include:

- Disease diagnosis
- Cancer status
- HIV status
- Medication
- Income
- Genetic information

A health dataset may contain several sensitive attributes simultaneously, which makes privacy protection more challenging.

8.3 Privacy Threats

Anonymization methods are designed to reduce different types of privacy threats.

8.3.1 Identity Disclosure

Identity disclosure occurs when an attacker can associate an anonymized record with a specific individual.

Anonymized record→Individual

8.3.2 Attribute Disclosure

Attribute disclosure occurs when an attacker learns sensitive information about an individual, even if the individual's identity is not directly revealed.

For example, an attacker may determine that a particular individual has cancer.

8.3.3 Membership Disclosure

Membership disclosure occurs when an attacker determines whether a particular individual is included in the dataset.

8.3.4 Linkage and Auxiliary-Information Attacks

An attacker may combine the anonymized dataset with external information to identify individuals or infer sensitive attributes. The paper emphasizes the importance of auxiliary information and adversarial knowledge when considering privacy risks.

8.4 Basic Anonymization Process

The paper describes a typical anonymization process consisting of several complementary stages.

Step 1: Removal of Direct Identifiers

Directly identifying information such as names, social security numbers, email addresses, and telephone numbers is removed.

Step 2: Selection of an Anonymization Technique

The appropriate technique depends on:

- data type
- privacy requirements
- intended use
- desired level of utility
- characteristics of potential attackers

Step 3: Selection of an Anonymization Operation

The original values or relationships are modified using operations such as:

- generalization
- suppression
- permutation
- perturbation
- anatomization

Step 4: Enforcement of Privacy and Utility Constraints

The anonymization process may impose constraints such as:

- minimum k
- acceptable information loss
- acceptable disclosure risk
- distributional constraints
- utility thresholds

The resulting dataset is:

$$D' = A(D)$$

where $A(\cdot)$ represents the anonymization procedure.

8.5 Main Anonymization Operations

8.5.1 Generalization

Generalization replaces a specific value with a broader category.

For example: Age=27 may be transformed into: Age=25–30 or: Age<30

Generalization reduces the precision of information while retaining some of its meaning. The paper notes that generalization can preserve more semantic information than suppression.

Relevance to causal inference. Generalization may reduce the resolution of important causal variables. Suppose age is a confounder: Age → Treatment and Age → Outcome

If age is generalized from an exact value to an age group, the information available for confounding adjustment is reduced. Therefore, an important causal question is: How much generalization can occur before causal adjustment is affected?

8.5.2 Suppression

Suppression removes or hides specific values.

For example:

Age ZIP

27 10115

* 10117

34 10119

Suppression can provide stronger privacy protection but may result in greater information loss.

Relevance to causal inference

If a suppressed variable is a confounder:

$X \rightarrow A$

and

$X \rightarrow Y$

removing or suppressing X may result in residual confounding and consequently affect causal effect estimation.

8.5.3 Permutation

Permutation involves shuffling values among records, often within defined groups. The purpose is to reduce the association between identifying information and sensitive attributes.

Relevance to causal inference. Permutation can also alter meaningful relationships between variables. For example, if $A \rightarrow Y$ is an important causal relationship, permutation of Y may weaken or destroy this relationship.

Therefore: Privacy protection $\uparrow$ may be accompanied by: Causal mechanism preservation $\downarrow$

8.5.4 Perturbation

Perturbation modifies original values or replaces them with altered values while attempting to retain useful characteristics of the data.

A simple example is: $X^* = X + \epsilon$, where ϵ represents noise. The resulting data may retain similar distributions while individual observations are modified.

Relevance to causal inference. Perturbation may change: $P(A|X)$ or $P(Y|A,X)$

even if: $P(X^*) \approx P(X)$

Thus, preservation of the marginal distribution of a variable does not necessarily imply preservation of the causal mechanism involving that variable.

8.5.5 Anatomization

Anatomization separates quasi-identifying information from sensitive attributes into different structures, thereby reducing the direct association between an individual's identifying characteristics and sensitive information.

The general objective is to retain useful information while reducing the possibility of directly linking an individual to a sensitive attribute.

8.6 Major Privacy Models

Four important privacy models discussed in the paper are:

1. k-anonymity
2. ℓ -diversity
3. t-closeness
4. Differential privacy

8.6.1 k-Anonymity

A dataset satisfies k-anonymity when every combination of quasi-identifiers is shared by at least k records.

For example, if: $k=3$, each individual should be indistinguishable from at least two other individuals based on the selected QIs.

The paper describes the re-identification probability under the basic k-anonymity assumption as approximately: $P(\text{re-identification}) = k^{-1}$

The primary purpose of k-anonymity is protection against identity disclosure.

Limitation of k-anonymity

k-anonymity does not necessarily prevent sensitive-attribute disclosure.

For example:

Age	Sex	ZIP	Disease
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55	F	10115	Cancer
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55	F	10115	Cancer
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55	F	10115	Cancer
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Although the three individuals are indistinguishable based on the QIs, their sensitive attribute is completely revealed.

8.6.2 ℓ -Diversity

ℓ -diversity extends k-anonymity by requiring multiple distinct sensitive values within an equivalence class.

For example:

Age	Sex	ZIP	Disease
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55	F	10115	Cancer
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55	F	10115	Diabetes
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55	F	10115	Hypertension
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This reduces the risk that membership in an equivalence class immediately reveals a sensitive attribute.

However, simply having multiple values does not necessarily mean that the distribution of sensitive information is adequately protected.

8.6.3 t-Closeness

t-closeness extends the idea of diversity by comparing the distribution of a sensitive attribute within an equivalence class with the distribution in the overall dataset.

Conceptually: $\text{Distance}[P(S), P(S | EC)] \leq t$

where:

- S is the sensitive attribute
- EC is an equivalence class
- t is the maximum allowed difference

This approach therefore considers the distribution of sensitive information rather than simply the number of distinct values.

8.6.4 Differential Privacy

Differential privacy provides a formal mathematical framework for limiting the influence of any single individual on the output of an analysis.

For neighboring datasets D and D' that differ by one individual's record:

$$P(M(D) \in S) \leq \epsilon P(M(D') \in S) + \delta$$

where:

- M is the privacy mechanism
- ϵ controls the privacy loss
- δ represents a small probability of failure

Generally: $\epsilon \downarrow \Rightarrow$ stronger privacy

Differential privacy is particularly important for this PhD because the privacy parameter ϵ provides a natural framework for investigating the relationship between privacy protection and utility loss. The paper identifies differential privacy as one of the major privacy-model families.

8.7 Evaluation of Privacy Protection

An anonymization method should be evaluated rather than assuming that it provides adequate protection.

Important privacy measures include::

Re-identification risk — $\text{Risky-id} = P(\text{correct identification})$

Attribute disclosure—How easily can an attacker infer sensitive information?

Membership disclosure—How easily can an attacker determine whether an individual is included in the dataset?

Adversarial evaluation—Can an attacker successfully link the anonymized data to external information?

The paper highlights accurate estimation of re-identification risk and modelling of adversaries as important unresolved challenges.

8.8 Statistical Utility Evaluation

Anonymization changes the original data, so the resulting loss of utility must also be assessed.

The paper discusses different utility measures, including information-loss measures, statistical measures, and application-specific utility measures.

Common statistical utility measures include:

- Mean
- Standard deviation
- Marginal distributions
- Correlations
- KL divergence
- Relative error
- Distributional similarity

Information-loss measures can include:

- Generalized information loss
- Discernability
- Minimal distortion
- Global loss measures

Predictive utility can be evaluated using:

- Accuracy
- Precision
- Recall
- F-measure
- Other predictive performance measures

These approaches determine whether anonymized data remain useful for statistical analysis or prediction.

8.9 Statistical Utility versus Causal Utility

A major limitation of conventional utility evaluation is that statistical similarity does not necessarily imply causal similarity.

Suppose the real data follow: $X \rightarrow A \rightarrow Y$, where X is a confounder, A is treatment, and Y is the outcome. An anonymization method may preserve: $P(X^*) \approx P(X)$ and even: $P(X^*, A^*, Y^*) \approx P(X, A, Y)$

However, this does not necessarily guarantee preservation of: $P(A^* | X^*) \approx P(A | X)$ or $P(Y^* | A^*, X^*) \approx P(Y | A, X)$

Most importantly, it does not necessarily guarantee: $P(Y^* | \text{do}(A^*), X^*) \approx P(Y | \text{do}(A), X)$

Therefore: Statistical similarity $\Rightarrow$ Causal similarity

8.10 Causal Utility of Anonymized Health Data

8.10.1 Covariate Preservation

Important confounders and covariates should retain their relevant properties: $P(X^*) \approx P(X)$

8.10.2 Treatment Assignment Preservation

The treatment assignment mechanism should be preserved: $P(A^* | X^*) \approx P(A | X)$

This is particularly important for propensity-score methods and inverse probability weighting.

8.10.3 Outcome Mechanism Preservation

The relationship between treatment, covariates, and outcome should remain similar: $P(Y^* | A^*, X^*) \approx P(Y | A, X)$

8.10.4 Causal Effect Preservation

The causal estimand estimated from the anonymized data should remain close to that from the original data:

$$ATE^* \approx ATE$$

Similar comparisons may be performed for:

- **ATT**
- **ATC**
- **CATE**
- **other causal estimands**

8.10.5 Causal Structure Preservation

The anonymization process should ideally preserve important causal relationships and mechanisms.

For example: $X \rightarrow A \rightarrow Y$, should not be substantially altered by the anonymization process.

8.11 Privacy, Statistical Utility and Causal Utility

The conventional anonymization framework focuses mainly on: $\text{Privacy} \leftrightarrow \text{Utility}$

For this PhD, the framework should be expanded to:
 $\text{Privacy} \leftrightarrow \text{Statistical Utility} \leftrightarrow \text{Causal Utility}$

Different anonymization methods may affect these dimensions differently.

Anonymization method	Privacy	Statistical utility	Potential causal impact
Generalization	Increased	Information loss	Loss of covariate resolution
Suppression	Increased	Information loss	Loss of confounders
Permutation	Increased	Variable	Disruption of associations
Perturbation	Increased	Variable	Alteration of causal mechanisms
k-anonymity	Increased	Variable	Possible change in causal relationships
ℓ-diversity	Increased	Variable	Possible alteration of sensitive-variable relationships

t-closeness	Increased	Distribution preserved	Distributional preservation may not ensure causal preservation
Differential privacy	Controlled through ϵ	Depends on privacy budget	Potential distortion of causal estimands

The important point is that the same anonymization method may have different consequences for different causal analyses.

Ref :

1. Murthy S, Bakar AA, Rahim FA, Ramli R. A comparative study of data anonymization techniques. In 2019 IEEE 5th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS) 2019 May 27 (pp. 306-309). IEEE.

2. Majeed A, Lee S. Anonymization techniques for privacy preserving data publishing: A comprehensive survey. IEEE access. 2020 Dec 18;9:8512-45.